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DANIEL LAHAV

CURRICULUM VITAE

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Department of Physics, Bar-Ilan University, Ramat-Gan, Israel | Citations: ORCID, Google Scholar

EDUCATION

Ph.D. in Physics, Bar-Ilan University, Ramat-Gan, Israel 2024 – Present
Thesis: "Flexible Spintronics Under Strain"

M.Sc. in Physics (Magna Cum Laude), Bar-Ilan University, Ramat-Gan, Israel 2022 – 2024
Thesis: "Flexible Planar Hall Effect Sensors Under Strain"

B.Sc. in Physics, Bar-Ilan University, Ramat-Gan, Israel 2019 – 2022

LIST OF PUBLICATIONS

1. Nhalil, H., **Lahav, D.**, Schultz, M., Amrusi, S., Grosz, A., Klein, L., "Flexible planar Hall effect sensor with sub-200 pT resolution", *Applied Physics Letters* 123(2), 024102 (2023). doi:10.1063/5.0156588

2. Lahav, D., Schultz, M., Amrusi, S., Grosz, A., Klein, L., "Planar Hall Effect Magnetic Sensors with Extended Field Range", *Sensors* 24(13), 4384 (2024). doi:10.3390/s24134384

3. Vizel, M., Alimi, R., **Lahav, D.**, Schultz, M., Grosz, A., Klein, L., "Magnetic Source Detection Using an Array of Planar Hall Effect Sensors and Machine Learning Algorithms", *Applied Sciences* 15(2), 964 (2025). doi:10.3390/app15020964

4. Lahav, D., Schultz, M., Grosz, A., Klein, L., "Enhanced Magnetic Resolution in Elliptical Planar Hall Effect Sensors via Non-Collinear Anisotropy Engineering", *Measurement* 294, 123683 (2027). doi:10.1016/j.measurement.2026.123683

5. Vizel, M., Klein, L., Schneider, N., Fisher, E., **Lahav, D.**, Grosz, A., Alimi, R., "Enhancing Super-Resolution Fidelity through Physics-Guided Multi-View Conditioning", *Array* (under review).

6. Lahav, D., Ghosh, A., Schultz, M., Grosz, A., Klein, L., "Shape anisotropy in patterned soft ferromagnets is constant only above a crossover field", *Advanced Electronic Materials* (under review).

7. Vizel, M., Alimi, R., Schultz, M., Fisher, E., **Lahav, D.**, Grosz, A., Klein, L., "Multi-Objective Optimization of Magnetic Anomaly Detection with Planar Hall Effect Sensors", *IEEE Sensors Journal* (under review).

8. Lahav, D., Schultz, M., Grosz, A., Klein, L., "Planar Hall Effect Sensor in the High-Field Regime: Sub-Microradian Resolution of the In-Plane Field Direction and Microtesla Resolution of the Perpendicular Field", *Sensors and Actuators Reports* (under review).

9. Forkosh, I., **Lahav, D.**, Schultz, M., Klein, L., "Detection of Subsurface Defects in Aluminum Plates with a Scanning Eddy-Current Testing System Based on a Single Planar Hall Effect Sensor", *NDT & E International* (under review).

10. Lahav, D., Schultz, M., Grosz, A., Klein, L., "Temperature-Drift Compensation of Planar Hall Effect Sensors from Their Longitudinal Resistance" (in preparation).

CONFERENCE PRESENTATIONS

Oral Presentations

- "Multifunctional High-Resolution Flexible Elliptical Planar Hall Effect Sensors for Magnetometry and Strain Sensing", XV European Magnetic Sensors and Actuators Conference (EMSA), 2026.
- "Flexible Spintronics Under Strain", KLA-BINA Outstanding Students Seminar, Bar-Ilan University, 2026.
- "Multi-Functional Flexible Planar Hall Effect Sensors", BINA Conference, Bar-Ilan University, 2025.
- "Multi-Functional Flexible Planar Hall Effect Sensors", The 16th Joint Conference on Magnetism and Magnetic Materials and Intermag, 2025.

Poster Presentations

- "Enhanced Magnetic Resolution in Elliptical Planar Hall Effect Sensors via Non-Collinear Anisotropy Engineering", XV European Magnetic Sensors and Actuators Conference (EMSA), 2026.
- "Expanding the Field Range of PHE Sensors for Increased Industrial Applicability", BINA Conference, Bar-Ilan University, 2025.
- "Expanding the Field Range of PHE Sensors for Increased Industrial Applicability", The 16th Joint Conference on Magnetism and Magnetic Materials and Intermag, 2025.
- "Multi-Functional Flexible Planar Hall Effect Sensors", International Symposium on Integrated Magnetics (iSIM), 2025.
- "Expanding the Field Range of PHE Sensors for Increased Industrial Applicability", International Symposium on Integrated Magnetics (iSIM), 2025.
- "Planar Hall Effect Magnetic Sensors with Extended Field Range", 69th Annual Meeting of the Israel Physical Society, 2024.

PRESS COVERAGE

"Flexible planar Hall effect sensor with sub-200 pT resolution":

- EurekAlert!
- The Jerusalem Post
- AZoSensors
- Israel National News
- Israel Noticias

FELLOWSHIPS, HONORS & AWARDS

Excel@BINA Scholarship for Outstanding Students 2026 – 2027
Institute of Nanotechnology and Advanced Materials, Bar-Ilan University.

KLA Scholarship for Outstanding Students 2026
Awarded for a lecture at the KLA-BINA Outstanding Students Seminar, Institute of Nanotechnology and Advanced Materials, Bar-Ilan University.

Excel@BINA Scholarship for Outstanding Students 2025 – 2026
Institute of Nanotechnology and Advanced Materials, Bar-Ilan University.

The President's Scholarship Program for Outstanding Doctoral Fellows 2024 – 2025
Bar-Ilan University, Ramat-Gan, Israel.

Excel@BINA Scholarship for Outstanding Students 2024 – 2025
Institute of Nanotechnology and Advanced Materials, Bar-Ilan University.

M.Sc. in Physics – Magna Cum Laude 2024
Bar-Ilan University, Ramat-Gan, Israel.

POSITIONS

Nano-Fabrication Center Operator, Bar-Ilan University 2023 – Present
Support cleanroom nanofabrication and characterization workflows for interdisciplinary research projects:

- Operate sputtering systems, lithography systems, and dicing equipment.
- Perform thin-film deposition, lithography, and device fabrication.
- Conduct AFM measurements in the MIXA characterization unit.

TEACHING EXPERIENCE

Teaching Assistant, Bar-Ilan University, Ramat-Gan, Israel 2022 – Present

Teach undergraduate laboratory courses:

- 86-251: Electronics Lab
- 86-232: Computerized Physics Lab
- 86-129: General Physics Lab
- 96-900: Preparatory Physics Lab

Tutor undergraduate course:

- 86-250: Electronics for Physicists

ACADEMIC SERVICE & OUTREACH

Student Representative, Physics Department – BIU Student Association 2023 – Present

Represent physics students in departmental and university committees and coordinate communication between students and faculty.

Invited Visit, HZDR, Dresden, Germany 2026

Worked with Denys Makarov's group on flexible magnetic sensors: shared our planar Hall effect and fabrication expertise, helped design a noise measurement system, and measured devices.

TECHNICAL SKILLS

Programming Languages: C++, MATLAB, Python
Software Tools: LabVIEW, COMSOL
Nanofabrication: Mask Aligner (MA), Maskless Aligner (MLA), Wafer Bonder, Ion-Beam Sputtering (IBS), BesTec (Sputtering and Evaporation), Disco Dicing Saw, ICP-RIE (Dry Etching)
Characterization: Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM)
CAD Tools: AutoCAD, Fusion 360, SolidWorks, CleWin
Systems Engineering: Design and integration of custom electrical, magnetic, and low-noise measurement systems, including automated experimental platforms